Regression, Inference, and Estimation Notes

1 Confidence and Prediction Intervals

1.1 Why CI Width Grows with Distance

For a new point x_0, the confidence or prediction interval involves the factor

$$1 + x_0^T (X^T X)^{-1} x_0,$$

because points farther from the cloud of training data have higher uncertainty. Intervals widen as you move away from the bulk of the data.

1.2 Estimating the Unknown Noise σ^2

Fit $\hat{w}$ by OLS and compute residuals $r = y - X\hat{w}$. Then

$$\hat{\sigma}^2 = \frac{\|r\|^2}{n-p}, \qquad n = \#\text{points},\ p = \#\text{parameters}.$$

Dividing by $n - p$ "pays" for each parameter we estimated, yielding an *unbiased* estimate of σ^2.

1.3 From Distribution to Interval

We know

$$y_0 - \hat{y}_0 \sim N\big(0,\ \sigma^2 s_0^2\big), \qquad s_0 = \sqrt{1 + x_0^T (X^T X)^{-1} x_0}.$$

Define

$$T = \frac{\hat{y}_0 - y_0}{\hat{\sigma}\, s_0} \sim t_{n-p}.$$

Using $P(-t^* \le T \le t^*) = 1 - \alpha$ and solving for y_0 gives the (two-sided) interval

$$y_0 \in \big[\ \hat{y}_0 \pm t^*\, \hat{\sigma}\, s_0\ \big].$$

1.4 Standardization Trick

Exactly like the simple normal case: from

$$P(-z^* \le Z \le z^*) \Rightarrow \mu \pm z^* \sigma,$$

we replace $Z \to T$ (Student t), $\mu \to 0$, $\sigma \to 1$ to obtain t-based intervals, then *undo* the standardization.

1.5 Shifting Normals

If $a - b \sim N(0, \sigma^2)$ with b fixed, then

$$a = (a - b) + b \sim N\big(b, \sigma^2\big).$$

Shifting by a constant adds to the mean, leaving the variance unchanged.

2 Variance and Standard Error of Coefficients

With $y \sim N(Xw, \sigma^2 I)$ and the OLS estimator

$$\hat{w} = (X^T X)^{-1} X^T y,$$

we have

$$\mathrm{Var}(\hat{w}) = \sigma^2 (X^T X)^{-1}, \qquad \mathrm{Var}(\hat{w}_j) = \sigma^2 \left[(X^T X)^{-1}\right]_{jj}.$$

Standard error:

$$\mathrm{SE}(\hat{w}_j) = \sigma \sqrt{\left[(X^T X)^{-1}\right]_{jj}},$$

and in practice we replace σ with $\hat{\sigma}$.

3 "Leverage" for Parameters vs. Predictions

- **Parameter SE** uses the jth diagonal entry of $(X^T X)^{-1}$.
- **Prediction PI** uses $1 + x_0^T (X^T X)^{-1} x_0$.

Both come from the same inverse, but you *slice out different entries* depending on whether you care about a coefficient or a prediction at x_0.

4 Error Model and Covariance

Assume the classical homoskedastic, independent errors:

$$\boldsymbol{\varepsilon} \sim \mathcal{N}\big(0, \, \sigma^2 I\big).$$

Interpretation for each error term:

(1) Gaussian (bell-shaped),

(2) Mean 0 (no systematic bias),

(3) Variance σ^2 (common noise level),

(4) Independence (covariances = 0).

Covariance matrix:

$$\mathrm{Cov}(\boldsymbol{\varepsilon}) = \sigma^2 \begin{bmatrix} 1 & 0 & \cdots & 0 \\ 0 & 1 & \cdots & 0 \\ \vdots & & \ddots & \vdots \\ 0 & 0 & \cdots & 1 \end{bmatrix}.$$

Plotting any two different errors yields a round, untilted cloud with an essentially flat best-fit line (slope ≈ 0): knowing one error tells you nothing about the other.

5 Maximum Likelihood Estimation (MLE)

5.1 Core Idea

Given data $x_1, \ldots, x_N \sim P(x \mid \theta)$,

$$L(\theta) = \prod_{i=1}^{N} P(x_i \mid \theta), \qquad \ell(\theta) = \log L(\theta) = \sum_{i=1}^{N} \log P(x_i \mid \theta),$$

and

$$\hat{\theta} = \arg\max_{\theta} \, \ell(\theta).$$

5.2 Why Log?

- Products $\to$ sums (simpler calculus),
- Numerical stability (avoid underflow).

5.3 Recipe

(1) Write down $P(x \mid \theta)$.

(2) Form $\ell(\theta) = \sum_i \log P(x_i \mid \theta)$.

(3) Differentiate w.r.t. each component of θ.

(4) Set derivatives to zero (normal equations).

(5) Solve for $\hat{\theta}$ and check it's a maximum (Hessian negative definite).

5.4 Worked Examples

- **Bernoulli (coin flip):** $\hat{p} = \dfrac{1}{N}\displaystyle\sum_{i=1}^{N} x_i$.
- **Gaussian, known variance:** $\hat{\mu} = \dfrac{1}{N}\displaystyle\sum_{i=1}^{N} x_i$.
- **Gaussian, unknown variance:** $\hat{\mu} = \bar{x}, \quad \hat{\sigma}^2 = \dfrac{1}{N}\displaystyle\sum_{i=1}^{N}(x_i - \bar{x})^2$.
- **Poisson (counts):** $\hat{\lambda} = \dfrac{1}{N}\displaystyle\sum_{i=1}^{N} k_i$.
- **Multinomial:** $\hat{\theta}_j = \dfrac{n_j}{N}$, with $\sum_j \theta_j = 1$.

5.5 When Closed Form Isn't an Option

Some models (e.g. logistic regression, mixture models) lack closed-form MLEs. Use numerical optimization:

- Gradient ascent,
- Newton–Raphson / quasi-Newton methods.

5.6 Practical Tips

- Always maximize the *log*-likelihood.
- Guard against $\log(0)$ (add tiny ε where needed).
- Regularization (priors) turns MLE into MAP, helpful for stability and controlling overfitting.

6 Sampling Distributions and the CLT

6.1 Definitions

- **Statistic:** function of sample data only (e.g. $\bar{X}$).
- **Population distribution:** distribution of a single X with mean μ, sd σ.
- **Sample distribution:** empirical distribution of one realized sample.
- **Sampling distribution:** distribution of a statistic across repeated samples.

6.2 Sampling Distribution of the Mean

For i.i.d. data with mean μ and sd σ:

$$\mu_{\bar{X}} = \mu, \qquad \sigma_{\bar{X}} = \frac{\sigma}{\sqrt{n}} \quad \text{(the standard error of the mean).}$$

6.3 Central Limit Theorem (CLT)

If $X_1, \ldots, X_n$ are i.i.d. with mean μ and finite variance σ^2, then

$$\bar{X} \approx N\Big(\mu, \frac{\sigma^2}{n}\Big) \quad \text{for sufficiently large } n \text{ (rule of thumb: } n \geq 30\text{).}$$

7 Z-Scores and the Standard Normal

7.1 Definition and Standardization

A z-score measures how many standard deviations x is from the mean:

$$z = \frac{x - \mu}{\sigma}.$$

Standardizing any normal $N(\mu, \sigma^2)$ yields $Z \sim N(0, 1)$ via center $(x - \mu)$ and rescale (divide by σ).

7.2 Standard Normal PDF and Empirical Rule

The standard normal pdf is

$$\phi(z) = \frac{1}{\sqrt{2\pi}} e^{-z^2/2}.$$

Empirical rule:

- $\approx 68\%$ within $\pm 1\sigma$,
- $\approx 95\%$ within $\pm 2\sigma$,
- $\approx 99.7\%$ within $\pm 3\sigma$.

7.3 Using Z-Tables

Tables give $P(Z \leq z)$. Then

- $P(Z > z) = 1 - P(Z \leq z)$,
- Two-tailed: $P(-z \leq Z \leq z) = P(Z \leq z) - P(Z \leq -z)$.

7.4 Examples

- IQ $= 130$, $\mu = 100$, $\sigma = 15$: $z = \frac{130-100}{15} = 2 \Rightarrow$ about 97.5th percentile.
- Height $X \sim N(170, 10^2)$: $P(X \leq 180) = P(Z \leq 1) \approx 0.8413$.

Bottom Line

- Always estimate the common noise level σ via residuals.
- For **coefficients**, the shape factor is $\sqrt{\big[(X^T X)^{-1}\big]_{jj}}$.
- For **new-point predictions**, it is $\sqrt{1 + x_0^T (X^T X)^{-1} x_0}$.
- Standardize $\rightarrow$ apply the appropriate t-quantile $\rightarrow$ undo the standardization.